

JUNIPERO VERBEKE

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Aerospace Engineering M.S. candidate (B.S. Applied Physics) passionate about **mechanical design and fabrication challenges**. Proven ability in **dynamic modeling** (Python, MATLAB), **control systems** implementation (FPV drones), and **flight data analysis**. Eager to contribute to engineering by leveraging **strong applied math skills, simulation and real-world hardware testing**. Based in San Jose, CA, with mobility to US and European regions.

EDUCATION

San Jose State University — <i>M.S. Aerospace Engineering</i>	Expected June 2027
University of California, Santa Cruz — <i>B.S. Applied Physics</i>	June 2025
Etoile du Matin - Eguelshardt, France — <i>French Baccalaureate - Mention très bien</i>	June 2020

COURSEWORK

Advanced Classical Mechanics, Feedback Control Systems, Linear Algebra, Vector Calculus, Differential Equations, Statistical Mechanics, Quantum Physics, Thermodynamics, Aerospace Structures and Dynamics, Python and MATLAB Data Analysis, Numerical Solutions, Simulation

EXPERIENCE

Custom Mechatronic Systems Design & Fabrication <i>Personal Business; CAD, Controls</i>	June 2020 – Present
- Designed, modeled (Blender, AutoCAD, Fusion360), and fabricated custom mechanical components for personal and customer projects using 3D printing (GCODE), CNC, and water jet cutting techniques. - Assembled, programmed, and tuned 4 FPV drones, designed for personal videography.	
Veritas Managed Solutions, inc. <i>Applied Engineer</i>	June 2021 – Present
- Planning and estimating 200+ security projects, including Tesla Gigafactories Sparks and Austin. - Spearheaded the formation of the engineering team, directly enabling new lines of business in security projects that contributed to Veritas's revenue growth in this sector from \$10M to \$28M. - Authored 5 universal guides for technician system implementation, decreasing time spent on projects by 5%. - Automated proposal creation using Word, Visio and Excel, decreasing time spent creating proposals by 80%. - Created 3D animations in Blender for publicity and introduced social media presence to the company, resulting in 3 new hires per month on average. - Renovated the company website, developing a full-stack web page in React increasing website visits from 24 to 4.4k per month.	
St. Thomas More School <i>Technology Support</i>	Sept. 2022 – Present
- Maintenance of St. Thomas More School's computer network and media rooms. - Filmed and photographed events with rapid turnaround (day of to 2 days from filming).	

PROJECTS

Physics-Based Numerical Modeling of Dynamic Systems <i>Python, NumPy, Matplotlib, MATLAB</i>	Nov. 2023
- Developed physics-based numerical models for complex dynamic systems from Lagrangian mechanics. - Analyzed computational efficiency of RK4 and Euler numerical solution techniques in Python and MATLAB.	
Aerodynamic Analysis and Fluid System Validation <i>Physics Lab (PHYS 134), XFLR5</i>	March 2025
Collected and analyzed experimental fluid system data, validating results against theoretical fluid dynamics principles and XFLR5 simulations to assess model accuracy.	

SKILLS

Programming Languages: Python, MATLAB
GNC & Technical Computing: Dynamic Modeling, Control Systems (Design, Analysis, PID Tuning), Linear Control Analysis, Simulation, Flight Data Analysis, Applied Mathematics (Linear Algebra, Vector Calculus, Differential Equations)
Software & Tools: Git/GitHub, Simulink, Fusion360, AutoCAD, Blender, VS Code, Unix Shell
Relevant Libraries: NumPy, Matplotlib, SciPy
Hardware & Fabrication: FPV Drone Design & Assembly, Flight Controller Configuration, Sensor Integration, 3D Printing, Water Jet Cutting